FICTION

Space rays

Jr. John W. Campbell

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SPACE RAYS

By John W. Campbell, Jr.

Illustration by Paul

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Reasonableness In Science Fiction

BY HUGO GERNSBACK

When science fiction first came into being, it was taken most seriously by all authors. In practically all instances, authors laid the basis of their stories upon a solid scientific foundation. If an author made a statement as to certain future instrumentalities, he usually found it advisable to adhere closely to the possibilities of science as it was then known.

Many modern science fiction authors have no such scruples. They do not hesitate to throw scientific plausibility overboard, and embark upon a policy of what I might call scientific magic, in other words, science that is neither plausible, nor possible. Indeed, it overlaps the fairy tale, and often goes the fairy tale one better.

This is a deplorable state of affairs, and one that I certainly believe should be avoided by all science fiction authors, if science fiction is to survive.

In the present offering, Mr. John W. Campbell, Jr., has no doubt realized this state of affairs and has proceeded in an earnest way to burlesque some of our rash authors to whom plausibility and possible science mean nothing. He pulls, magician-like, all sorts of impossible rays from his silk hat, much as a magician extracts rabbits. There is no situation that cannot easily be overcome by some sort of a preposterous scientific--(as he terms it)--gimmick. Hurling fifteen million horsepower from one space flyer to another means nothing. If he has left out any colored rays, or any magical rays that could not immediately perform certain miraculous wonders, we are not aware of this short-coming in his story.

In other words, the author proceeds to burlesque science fiction--not only science fiction, but he burlesques his own abilities, to show us what really can be accomplished when the bounds of reasonableness are overstepped.

Yet, all in all, he has spun a delightful tale which will, nevertheless, entertain you, and keep your interest throughout. We were tempted to rename the story "Ray! Ray!" but thought better of it.

I have gone to this length to preach a sermon in the hope that misguided authors will see the light, and hereafter stick to science as it is known, or as it may reasonably develop in the future.

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Eight large men were crowded in a small emergency space ship. They were physically crowded; the air was bad for the little ship was built for three men, and the capacity of the air apparatus was hardly sufficient for them. And so the E.E. 4 was wobbling under her controls, with acceleration greatly reduced.

Lt. Roger Lane and his men felt exceedingly foolish. They had just surrendered without protest to a pirate, for the altogether-too-good ship _Robin Good_ had suddenly appeared alongside the Lane Patrol ship L.P.C. 563. The air in the patrol cruiser glowed faintly blue, and the crew of seven men and two officers slumped down where they were. The latest Farrel Atomics, capable of generating fifteen million horsepower for the defense of the cruiser hadn't even started. Nine helpless men watched while _Robin Good_ sent over a shipload of men in space suits. Each carried a small hand projector, and without entering the beam from the ship they turned their small projectors on the men in the cruiser.

Fully conscious, but helplessly paralyzed, Lt. Lane and his men were dumped unceremoniously into this tiny escape ship, and turned adrift, without radio or weapons. Lane was heading with all speed for Mars, some thirty million miles away. But one of the pirate crew had thoughtfully removed one of the miniature Farrel Atomic engines, with the result that their McKinley space-drive plates were not getting the power they should have--the lights were dim, and the already over-loaded air apparatus was functioning badly.

Pilot Warren was in the tiny cubby-hole in the nose of the ship, speeding with all the acceleration he could muster for Mars, although the officers of the ship were not so enthusiastic about getting there. It would have been bad enough to have been unable to find the pirate after being sent out for the purpose of overhauling his slower ex-space-yacht, but to have lost the speedy, powerfully armed space cruiser without a struggle! Commander Morrison would make some exceedingly pointed remarks, and it would be a long, long time before Lane got a new rating.

"I'll bet Barclay gets a work-out outa this," said Patrolman Reynolds, sourly.

Yes, decided Lane, Special-Captain Donald Barclay would probably get some work. He hoped that Barclay would be picked up by the pirate too. Barclay had an annoying habit of doing what he was sent to do. Nobody ever said anything about it, least of all Barclay, but those reports on the bulletin board had become monotonous.

Lt. Lane permitted himself to grouse heartily, silently and hungrily for thirty hours while the escape ship limped into port. He decided to permit himself a meal and a shower before appearing before Commander Morrison. It would be a televiewappearance, because Morrison was on Earth, but the Commander had long since learned the knack of being nasty over the televisor. He would be specially nasty this time. Lane took a long pull on some forbidden _tekey_--fermented Martian cactus juice--and felt ready to face the Commander.

Half an hour later he came out of the booth. It was after sunset, the air was chill, but Lane was sweating profusely. Morrison had just grinned and nodded. Lane wished he could grin like that. Then he wouldn't have to bawl out sloppy workers. Lane went to the barracks and was soon asleep.

Meanwhile Commander Morrison started things moving. As he cut off the televiewimage of the harassed young officer on Mars, he pushed a button, and his orderlyappeared.

"Special-Captain Barclay, at once please." The orderly saluted smartly andright-about faced. Ten minutes later a peculiar looking man entered the room behindthe orderly. Morrison greeted him with a friendly, easy salute.

Special-Captain Don Barclay smiled at him. Barclay looked to be about five feet tallby four feet wide. Actually he was nearly five-feet-seven, but his enormous breadthbelied it. Barclay was a type of man seen rather rarely on the minor planets. He was aJovian.

Three generations back his people had migrated from Earth, when Jupiter'scolonies were newly opened, in 2250. Men and women picked by the InterplanetaryCouncil, for both physical strength and health, for courage and for trustworthiness.To them had been given the task of developing a new race, a race of giants. Under agravitational force of from two and one half to three times that of the Earth, they hadto learn to live again. The cowards never started, the weaklings died. Never was itmore true than of this Jovian colonization.

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Born and raised on this planet, where all things must move rapidly to keep up withtheir surroundings, where a falling body moved two and a half times as quickly as onEarth, Don Barclay had become a giant. His legs were thick, heavy and almost blocky,his arms were built on the same plan, his tremendous shoulders seemed rounded andpadded as though he wore wadding under the tight elastic Patrol uniform. Hismassive head, crowned with crisp black hair, seemed set directly on his shoulders,the neck seemed missing. Five-feet-seven--and he weighed over two hundred andninety pounds!

"More work, Commander? I thought I was to have a week off after that little run-inwith the Eugenic people." He smiled, his lean, almost black face, burned fromceaseless exposure to the unshielded ultraviolet rays of the sun, creasing in a good-natured smile.

"Uhuh--so did I. But remember, this isn't a business, it's a war, only we don't knowwhom we're fighting half the time. This time it's Percy DeWare Jennings, betterknown as 'Robin Good'," replied the officer sourly. "He just stole one of our ships--theL.P.C. 563, Lt. Lane in command."

"Ouch--he tickled us that time. That does demand some action. What else has hebeen doing. I don't know all his story."

"Started off with a ten-man ship he bought second-hand on Mars, only half amillion horse, but he and his gang captured an ex-yacht converted to an excursionship. They used that about a month, then they captured this ship of ours. Fifteenmillion horse in perfectly good Farrel atomics, with induction rays, magnetic andinduction screens, super-sensitive electron balance detectors. It hasn't gotprotolectric guns, but has got the shields. In other words he's got everything we've

got, plus his own paralyzing ray. All in all a sweet situ--"

Commander Morrison stopped. His orderly had intruded hurriedly with "Pardon me, Commander, but this broadcast--" He snapped on the news broadcast machine, and the screen glowed with the image of a popular humorous news man.

"--anonymous person has sent in this tip from space. They gave data, and checking shows it's right. Believe it or not, 'Robin Good' has sneaked one over on the Patrol for sure this time. He stole their patrol Cruiser 563, got away with it, and sent the men home, neatly wrapped up in an escape ship designed for three men. They landed on Mars, I hear. All this happened yesterday, Sol 3, of the year of our Lord 2374. I certainly would like to have seen our good Commander Morrison's face when Lt. Lane, commanding 563 reported _that_ in. Hah! Hah!"

"Damn! Now who in the name of the nine planets sent that dope to that blabber-mouthed nit-wit!" roared Morrison. "I'd like to twist his squawking neck for--"

"And," continued the news machine imperturbably, "I'd like to see Commander Morrison's face right now if he's listening."

Had he been able to, he would have seen it tinged with an apoplectic purple, the stiff, iron-grey hair bristling erect on his head, his mouth open and gasping for words to describe his opinion of the man.

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The orderly shut off the machine and hurried from the room before Commander Morrison could regain his tongue.

When the color had receded, his mouth closed with a snap. He turned on Barclay with a roar. "Captain, GET THAT MAN, AND BRING HIM IN ALIVE!"

"I'll try," said Barclay, saluting. He left the office. "Whew--I think I'd better. And it'll be one sweet job. Nobody ever sees that man's ship; he's supposed to have found the secret of invisibility, and I'm supposed to find him.

"I think some research is in order." His stride suddenly became purposeful as he started for the elevator. He dropped swiftly to the street, went down the hill to the edge of the Patrol Headquarters grounds, and entered the city of Morristown itself. At the gates he paused long enough to slip on a suit of civilian clothes. Then he went directly to the monorail station, and swung aboard a car bound for New York City 33rd. The streamlined car darted forward as a McKinley drive plate in the nose came to life, and hurled itself along the rail fifty feet above the countryside toward the city.

At 33rd Station he descended, and stepped aboard the uptown walk, quickly moving over to the medium speed lane. At thirty-five miles an hour he was swept along through the magnificent canyon of New York. The flashing vitreous walls of huge buildings, made of colored glass bricks, windowless, save on the first two stories where show windows broke the sweeping lines of the pinnacles and lighted the street with reflected lights of red and gold and blue. Within, perfectly controlled artificial lights and perfectly comfortable artificial weather displaced the glaring sunlight and humid heat of the streets.

To the man of Jupiter these cities were forever wonderful. Jupiter, burdened with its heavy gravity, could not build so high, because the sheer weight would have pulled the structure down. On Mars was the Martian Lines Tower, carrying the flag of the Martian Lines interplanetary ships one full mile into the clear, crisp air--standing tall, almost alone among the massed ranks of giant structures that made this a city like none other.

Finally, at 87th Street he left the moving walk, and went down to the towering skyscraper library of the City of New York. It was the scientific news journals that he inspected first. Journals of six and eight months back, before Dr. Percy DeWare Jennings had gone a-pirating, when he was a highly respected bio-physicist.

But there was little to indicate that he might have discovered the secret of invisibility, his work had not been along those lines. But six months ago he had announced his discovery of the long-sought paralyzing ray that was not fatal. Others had made paralyzing rays that paralyzed--permanently, however. The trouble was that they paralyzed the heart and the lungs as well as the arms and legs. But Jennings had announced his discovery of a non-fatal paralyzing ray. Scientists of the world had scoffed--it was an impossibility. "Come and see!" demanded Jennings.

They came. But they did not see, for something went wrong with the apparatus. First it killed a dog, then a mouse ran about under it unhampered. The scientists went home laughing.

Perhaps Jennings had gone slightly mad. At any rate he disappeared for five months, to turn up with a pirate gang. They looted passenger liners, helpless under the paralyzing ray, but they themselves could not enter the ray without being paralyzed, for they had no screen for it.

Don Barclay mused, investigated, and got later data--accounts of the experiment written by scientists who had seen it, both immediately after the demonstration, and five months later when Jennings proved his claims all too thoroughly.

It was a long patient search, and took two days. The third day Special Captain Barclay of the Interplanetary Lane Patrol became Dr. Donald Barclay, and went to the Patrol research laboratories. He spent three more days there. Commander Morrison asked him why he hadn't started, and Barclay merely replied he had gone.

But on the sixth day he reported to Commander Morrison, and asked to be assigned to a ship.

"Naturally," snapped Morrison. "But why didn't you come after it sooner?"

"I was working on leads here," replied Barclay. "And may I have X-531?"

"What!" gasped Morrison, "X-531? That's a one-man experimental ship."

"With ten million horse in the brand-new Ridgely Automatic Atomics," replied Barclay. "And I want another ship anyway, the new Twenty-five-Hundred. And I'd like Dave Wier to take it, with a crew of Jovians. I'll try to find the pirate myself in the small ship. I can hide a mosquito like that in space, with a little black paint on it, and some red windows, with green lights in the ship. I'm betting that's what this Jennings

does with his _Robin Good_ anyway. I hear that Grantland has just finished a new type fast-action infra-heat photocell that will work a viso-screen. I want one of those installed, that'll make anything visible in space. And--here's something to install in the I.P.C. 2500."

CHAPTER II

Barclay Goes to Work

Morrison looked at his crack Special-Captain sourly.

"Oh, yes sir, and can I give you the rest of the fleet?"

"Will you if I bring in Jennings and his crew--unharmed?" smiled Barclay, looking at the news-receiver, clucking and chuckling to itself on low power.

Morrison got slightly red and snapped "Yes!"

"And look at this, sir!" suggested Barclay. Morrison took the sheaf of papers, and glanced hurriedly at the first lines, stiffened, and scanned the paper rapidly. "Good Lord, will it work?"

"It does, Commander, I tried it. I did some research on reports I found in the Public Library, and made some experiments."

"But--don't you want it in the X-531?" he asked in surprise.

"No sir, I don't. I want the Miller Twins instead. I have an idea that works better this way."

"But the pirate may capture you, and learn the secret of the Miller Twins. That's our latest and best weapon. Why, the Twenty-five-Hundred isn't even commissioned yet, and she's the only one that's got them, and the insulation," objected the Commander. "I can't let Wier take that ship, she isn't commissioned."

"Commander, I'll bring in Jennings if you'll let Wier take that--if he has to steal it unofficially. Why can't Wier take it out on trial maneuvers?"

"Ahh--well," said Morrison, scowling at the news machine, "go ahead--but if that half-witted flap-eared donkey reports that Jennings is using the Miller Twins before any commissioned ship--you needn't report back!"

Barclay's lips cracked in a slight smile. "He may--but don't worry Commander--it'll be my own hide that suffers."

Barclay left with the requisition slips Morrison had been filling out. And twenty-four hours later, after fifteen hours of concentrated sleep, Barclay left Earth in a tiny pea-pod of a ship, scarcely twenty feet long. But it carried ten million horsepower generators humming with a soft, pleasant melody behind him. They occupied almost the entire bulk of the ship, the pilot was forced to sleep in his pneumatic springed pilot chair.

A million miles from Earth, on his way toward Mars, Barclay took a small key and unlocked the acceleration stop-block. Presently he moved the metal stop-block from its position at "4000" to "7500". Four thousand centimeters per second per second was all any Earth-native could stand in the way of acceleration and still retain sufficient consciousness for fighting. Hence the lock, lest some foolhardy fighter try more. But Barclay, brought up on a world where the natural acceleration and gravity was nearer 3000, could readily stand the higher value.

And somewhere in space he knew that Dave Wier, his support in the Twenty-five-Hundred, was manipulating a similar key in a similar lock-block. His crew of twenty-two men were all Jovians, for that acceleration might well be fatal to the Earth-native.

Presently Barclay locked the block again, closed down the control panel hood, and forced the accelerator control over. He pushed steadily till his face grew pale under the nearly black complexion years in space had given him. His arm was knotted with great muscles, the shoulders bulging and straining at the fabric of his uniform. A black metal needle had long since passed the red danger line on the accelerometer gauge, and hung at 7500. Barclay gradually eased back, and the needle dropped to a comfortable 2750. Barclay got up, and stretched under Jupiter-normal.

He prepared a hasty meal while the ship darted along under automatic controls, jogging slightly now and then as a meteor came into the range of the exceedingly sensitive electro-static mass-detectors. Presently he finished, cleaned away the scraps by dumping them into space, re-seated himself and shoved the acceleration up a bit. Hour after hour he bored on into space, his ship a black dot in a black void.

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Three days later found the tiny ship far beyond Mars, circling near the asteroids. _Robin Good_ was reported to work out this way in search of Jupiter traffic. Barclay had been following the shifting Jupiter-Earth lane for two days now. The Space Liner _Mira_ was due to pass in another day, from Jupiter to Earth, and loaded with the usual assortment of heavy metals, platinum, iridium, osmium, some uranium, and a full pound and a half of jovium, element 97, on which the pirate must depend for renewing his atomic engines.

The space patrol cruiser had had only a sixteen-hundredth of a gram left in the supply bins, and an eight-thousandth of a gram in the engines. This might well last a year in ordinary service, but under the heavy drains the pirate's constant warfare would make, it would probably give out in six months. With a pound and a half at his command, he could retain a full ounce, a lifetime supply, and sell the rest of the activator. Barclay felt sure he would attack.

Barclay was right. He was sleeping when the exceedingly delicate electro-static apparatus gave him the first warning of the _Robin Good's_ approach. The tiny X-531 spun on her axes in an instant--and space was empty. Nowhere could Barclay get a sight of the ship the instruments said was less than a mile away, and approaching! Then--the infra-red screen, and the long, black hull was suddenly visible exactly where the instruments had said it would be. Invisible in ordinary light, the radiated

heat made it shine a pale green on the screen.

Chuckling, Barclay shoved home a small control lever. The white cabin lights suddenly snapped out, and green lights came on in their stead. Simultaneously, the ship leapt forward at her full 7500 acceleration, at right angles to the line between the two ships. Outside, the pirate looked in vain for the tiny ship that had been shining red before him. The green light did not penetrate the red glass windows. Barclay was lost, and the special electro-static field he was setting up, fluctuating wildly, made it impossible to locate him with the instruments.

Ten miles from his original position, he swung, till the nose of his ship pointed toward the pirate craft, and a pair of hair lines crossing on the center of the infra-screen covered the center of the pirate ship. The X-531 suddenly chattered violently as a machine gun, shooting one inch slugs loaded with atomic explosive iron dust, raked the pirate craft from stem to stern.

The ex-cruiser's magnetic screen stopped them, but they resulted in a magnificent display of fire-works that would have blown an entire nation off the face of the Earth. Barclay moved. From another angle he centered his cross-hairs on the screen and pushed a button. Suddenly a spot five feet across on the wall of the pirate ship was blazing red--orange--and then black, while a sea of splashing, warring energy appeared on the induction screen.

The arcing decreased abruptly as Barclay cut down the power on this, and turned five million horsepower into an infra-heat ray. The result was an instant blaze of white light, and a puff of white-hot vapor from the bow of the _Robin Good_. The black outer-hull soaked up the heat like a blotter taking water--and destroyed itself. But suddenly, in the wink of an eye, the black was gone, and the clean, shining metal of the patrol cruiser appeared. Instantly the reflected heat struck the little X-531 and Barclay was compelled to shut off his beam and shift hurriedly.

A sizzling induction beam from the pirate snapped by where he had been a moment before. Barclay felt it was time for the Twins.

Twin, milky-white columns of light reached out like pipes of milky glass. They swung abruptly, and touched the pirate ship some twenty-feet from the tail. Instantly the metal between the two beams blazed into white heat, and exploded out into space. And immediately after, a second puff from the same spot indicated the destruction of the inner skin of the ship. A long fountain of light stood out from the wall of the ship as the air rushed out.

Barclay moved his beams slowly and steadily, while the generators behind him whined and howled steadily. Ten million horsepower were pouring into those beams, and into the metal of the other ship. For less than five seconds that work continued. The _Robin Good_ rolled, twisted and struggled, but the little mosquito ship hung grimly in place, and poured its deadly beams in. Then a pale blue-green ray, scarcely visible in space, reached back along the twin milky beams, and touched the mosquito ship. Suddenly the X-531 followed a perfectly straight course, and the twin beams wandered far afield.

Inside the ship the air became luminous with the blue-green beam, and Barclay slumped inertly forward over his instrument panel. His eyes were open, but stared motionlessly straight ahead. Don Barclay was fully conscious, but absolutely unable to move a muscle.

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The blue-green luminosity crept forward in the ship, till only the pilot and the pilot's seat was bathed in it. Presently something grated on the emergency escape air-lock, and there was a hiss of gas, then a steady pumping. A moment later a giant of a man stepped from the inner lock-door, a flashlight in one hand, a ray-pistol of some sort in the other. It was a curious mechanism, a pistol grip, a flexible lead looping back along his arm to a small pack on his back. The business end of the pistol ended in a lens of wires, woven and crossed with infinite care, the whole protected by bars of metalite, an insulating, non-conductor of heat with the strength and polish of many metals.

The man was a giant, but such a giant as one might see in dreams--nearly six-feet-eight in height, broad in proportion. He wore only light green rayon shorts, a short-sleeved shirt to match, and a broad leather belt from which hung three holsters, and two more pistols.

But his face was an awful travesty of what Nature had meant. At some time he had been in a ray-fight, the entire left side of his face was a mottled purple and angry red, the right side was puckered and twisted till it drew the flesh away from the teeth, a horrible artificial grin resulting. The tip of the nose was entirely missing. But from just below the eyes to the hairline the flesh was smooth, and deeply tanned, almost black from exposure to the sun of space. The face was topped by a shaggy, unkempt mop of glaring red hair.

The motionless Barclay groaned inwardly. Belasco Red! Six months before Lt. James McRuddy had reported the death of this devil in human form. When the Belasco gang had been cleaned out of Thenton, Mars, he had been rayed, across the face, the flesh charred black. A companion had been there to drag him away. The companion had been captured two days later and told the Patrol that Belasco Red had died and been cast into space, while the ship struggled to make good its escape.

Belasco Red was grinning more broadly now. The pistol suddenly shot out a blue-green ray which centered on Barclay's neck. A great hand waved, and the beam from the pirate ship snapped out. Belasco walked forward gingerly, keeping the ray on Barclay, then picked him up and carried him back to the lock, through the metal communication tube and into the pirate ship.

"Looka wat we caught," he laughed, dumping Barclay on the floor. For a fraction of a second the patrolman was out of the ray, and the instant his body touched the floor he leaped up like an un-leashed spring. His hand darted toward the giant outlaw's pistol, but halfway it fell to his side as the ray touched his neck once more.

"It wants to fight I think," said Belasco. "It's a patrolman from the looks of the suit. Shall we play with it awhile?" His grotesque face writhed into an expression

impossible to describe.

A man of medium height, with smooth sandy hair had looked into the room, where three other pirates had been when Belasco returned.

"Oh, I say, that looks like Special-Captain Barclay. Truly we're honored, you know, Captain, but we must detain you a while.

"And, Red, let's let him help us a bit. I'm sure the fellow will be glad to. That was a nasty little jolt he gave us with those skim-milk rays of his. Did you look at the apparatus, Thad?" He spoke to a stockily-built man to one side. Thad was nursing a burned hand.

"Yeah--it's been gimmicked. H-T stuff. Almost took my hand off. I was careful enough, but the damn thing turned loose half a million volts soon as I moved the control panel. I'll be damfino how it could. I shut off the generators. They were new, too, but they blew out when the beam-control went off."

"Ah--the new Ridgely Automatic Atomics. They can start themselves, Thad. And they're much more powerful. Too bad. We'll have to capture the Twenty-five-Hundred when they commission that. I hear it will be equipped with the new generators.

"Well, it's all right. Cast the machine loose, burn it, and I'm sure Special-Captain Barclay can help us. I know a bit about rays and such myself. We should be able to figure it out together, don't you think so Captain?" The lean man looked at Barclay. "Yes, I think so.

"By the way, Red, just go on and move into position for the _Mira_. We won't need to make repairs before then, and haven't time. We don't know just when she'll come, it may be a day or two, and it may be at once. We can make repairs there."

The disfigured giant gave the ray-pistol to the stocky Thad, and departed on his errand. Presently the ship swung somewhat in its course.

"Thad," said the sandy-haired one, "suppose I try using that pistol. I'd like to talk to our friend here, and he seems sulky." Silently the blocky Thad passed the weapon over. Quickly the new possessor made several adjustments, then moved the beam from the neck to a point just below the middle of the chest. Don found his neck and head as well as his eyes could move now.

"Hello, Jennings. That feels better."

"Ah--quite so, quite so. Have you heard of the stimulator, I wonder?"

"No, I admit ignorance of that. A new invention?"

"Yes--quite. We ought to try it out, don't you think so, Thad? Please bring me the portable set." Thad disappeared silently. Barclay looked at the green-clad man before him. Save for the bright red-leather belt at his waist, his uniform was exactly like the others.

"Is it Lincoln green?" asked Barclay. "I never did know colors."

"No," replied Jennings with mock sadness, "it isn't. We couldn't get it. It isn't made, and I haven't had time to make any. Managing this little adventure is really trying. Belasco is so insistent that we remove all of your confrères on sight, he becomes hard to control at times. Ah, the Americans are so enterprising. He wants to destroy the entire patrol. I'm more modest in my desires. We English are apt to be. I'm planning to induce certain so-called scientists to witness a demonstration. That will probably take some time, and I do hope I will continue to have Belasco's co-operation. He's almost impossible.

"You see I treated his face, and not being a skilled plastic surgeon, it isn't beautiful. But he lived. It's healed now. I used a certain drug during the process, very quick healing, but unfortunately it's habit forming. He has hemorrhages of a most frightful sort if he doesn't get his daily ration. And I'm the only one who can make it.

"He tried once to make me give him the secret. Just once. But he's still hard to handle sometimes. You for instance. He will insist on being unpleasant about it. And I am hoping to take certain captives from the _Mira_ this time, with our larger ship, equipped as it is with a brig for prisoners, it should be a lucrative trade. But he will probably be unpleasant.

"Ah, Thad. Thanks. This is the stimulator, Captain. Stimulates certain nerves. See?"

CHAPTER III

Man and Woman

A pale orange ray reached out of the cone-shaped projector, and touched Barclay's foot. Involuntarily a cry rang from him. It felt like nothing describable, it seemed a red-hot vise was clamping on it, crushing every bone and tendon, while a dozen rats with white-hot teeth gnawed at it. It lasted but an instant, then the orange ray snapped out.

"The stimulator," said the bland Jennings, "affects only the nerves leading to the brain, stimulating them instead of paralyzing the nerves leading from the brain as does the paralyzing ray. It affects all the nerves, so the sensations are mixed, a feeling of fire, of crushing pressure, of breaking bones, itching, everything is stimulated.

"Very interesting," Jennings nodded to himself. "Very effective," he smiled.

A muffled vibration from the beryllium metal flooring announced the approach of some massive man, and presently Belasco's ghastly mask of a face appeared.

"Ah, entertaining him? Does he know what that apparatus is all about?"

"Certainly. He probably aided in its construction. They wouldn't send a man who knew nothing about it to test it would they?" snapped Jennings. "You do, do you not, Captain?"

"No," lied Barclay immediately. "It was secret stuff; they told me how to work it, rigged it with that gimmick, and sent me out. I couldn't even turn the gimmick off."

"There. You see he knows. Any man who didn't wouldn't be so explicit in what he didn't know, but would tell all he did," said Jennings, turning back to Belasco. "In the meantime, let's get that apparatus installed. You will show us how, Captain?"

"No, I can't, anyway," snapped the Patrol Captain.

Belasco's ham-like hand took the cone of the stimulator projector and a broad grin appeared on his face. "No?" The ray touched Barclay's foot. For a moment he stood it grimly, but as the ray deepened in color, as more power came on, he groaned aloud, sweat started from his forehead--and yet the ray was not actually harming him! Belasco grinned wider, and increased the power. This was not pain that was caused in a natural manner, it was pain that was actually a harmless nerve stimulation! There was no destroying fire to deaden the nerves, and lessen the agony, no grinding vise crushed them. They were merely stimulated.

In thirty seconds Barclay was willing to tell them what he knew. Belasco kept it up for a full minute. "More," he promised, "if we don't like your behavior. Will you need anything we won't have on the ship?"

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The pain left the instant the ray snapped off, and Barclay sat up dazedly, leaning back against his hands. His legs were powerless, a pencil beam shone on them, paralyzing him from the waist down. "Yes," he said faintly, "if you've got the regular stock-supplies."

"And more, Captain," Jennings assured him, "we always sail well equipped, as I like to do experimental work."

"All right. You, and your ray-engineer," said Barclay, "will have to help. And you'll need some extra heavy conductor cables. To be effective you've got to turn every horsepower of your generator into the Twins. Twin beams. And don't put fuses in the lines--" He talked rapidly, explaining the apparatus, how it projected a beam that would carry a current of electricity across empty space, explained that if the beams carrying their millions of horsepower encountered a rib in the ship, a surge might develop that would blow out a fuse, without reason but with disastrous results in battle, for several minutes would be needed to make replacements.

Belasco hung in the background as they worked at plans and drawings and calculations, his paralyzer and stimulator projectors ever ready. A fixed projector had been arranged that kept Barclay helplessly pinned in his seat, his legs powerless.

The ship rang with the clangor of repair workers at work on the damage done by the X-531.

In some four hours Barclay had finished the plans. Then he was promptly removed, and put in the brig, stout chains fastening him to the metal wall. The room, about thirty feet long by fifteen wide, was ordinarily used by the Patrol as an auxiliary storeroom, but the stores had been removed in preparation for the expected

prisoners of the _Mira_. One tiny porthole opened on space, to give a view of star-studded heavens.

A row of bright lights down the roof offered illumination. The door was a single slab of solid steel three inches thick, sliding vertically against the ship's normal acceleration, and held in place, when locked, by the series of tremendous magnets all about it. Once settled in place, nothing short of a disintegration ray could move it. To move the door in any way, to fuse it, would generate a terrific current in it, and automatically give the alarm, as well as stun anyone standing on the metal plates of the brig.

Fifteen feet from the floor was the roof, and five feet below the roof was a little platform on which a guard could sit to watch a number of prisoners.

Belasco kicked Barclay in, and left with a growl. He would have preferred to kill the Special-Captain then, but Jennings pointed out that the projectors might not work after all.

He was securely chained, and the men left, locking the door magnetically. Barclay disposed himself on the hard bunk, stoically, smiled resignedly, with a flash of humor in his eyes. He went peacefully to sleep.

He woke, hours later, to the clang of the great metal door as it was shot upward in its slides. A small wiry man, clad in the green uniform of the pirate ship, entered and stepped to one side of the door, a sly grin of pleasure on his face. There was a portable stimulator set in his hand, which he set on the floor, retaining the projector cone and control mechanism in his left hand, and a paralyzer in his right. A moment later there was a heavy trampling on the metal floor, groans and low voices, a sharp command to "step along" in Belasco's harsh voice, and a string of men entered the room, clad in civilian clothes. There were tall ones and short ones and thin and fat, but they were all marched in and fastened by chains to the bunks.

Then finally two women came. The men had been reacting to their capture in all ways, from the height of indignation to the depths of groveling fear. The two women marched proudly in, looked over the room with a swift glance, and marched over to the least-crowded corner, about twenty feet from Barclay.

One was young, about twenty-two, the other, evidently her mother, about forty-five. Both were tall, broad and heavily built. The daughter was Junoesque, if not more. Her mass must have been easily 170 pounds, and her mother was equally heavy.

And, bringing up the procession, came Belasco. He entered the doorway, and stood waiting till everyone had been chained, his hideous face distorted in what was supposed to be a smile of satisfaction.

Silently, a second figure appeared behind him--Jennings. Quick dark eyes scanned the prisoners, and with a sudden narrowing stopped on a tall, powerfully built civilian, his face a deep brown from long exposure to Martian sunlight. But he was a terrestrial, his smooth-fitting elastic shirt proclaimed his muscles were well developed, not as a terrestrial becomes after months on Mars.

Jennings passed Belasco, and stepped nearer to the man. "Who," he demanded, "are you?" still scrutinizing him closely.

"Jimmy Durante, terrestrial engineer, working on Mars. Just made a trip to Jupiter for some special machinery when your murdering gang of cut-throats killed off the crew of the ship, and took us. I hope you get blood-poisoning."

* * * * *

Jennings' face had relaxed from its intense stare of concentration. "No," he smiled, "you aren't Jimmy Durante, you're John Norcross, and you're Special-Lieutenant John Norcross at that. You take the air. We don't want you." He pulled out the red-butted pistol that hung on his right side, aimed it deliberately at the center of the Patrol Investigator's forehead and pushed the trigger button. A blinding flash of white light sprang from it, a momentary odor of burned flesh, and the light disappeared as Norcross dropped.

"Take that," said Jennings to another uniformed pirate, "and throw it down the waste-chute." Jennings turned, and started back toward Belasco. "I'm going up to the pilot-house. The Twins have been finished, and while you were tying up our friends here, I tried it out on the old _Mira_. Cut it in two nicely."

"Shall I finish off this thing now?" asked Belasco eagerly, looking toward Barclay.

"Hmmm--no, I wouldn't yet. I...."

"I wouldn't either, Jennings," said Barclay, "I saw a piece of apparatus back at H.Q. that might interest you. Shield for the paralyzer ray, they claimed."

"That," said Jennings judicially, "is a lie. There isn't any. I can't make one myself, and I made the apparatus, and have the only ray apparatus."

"Of course you are also the best physicist in the system?" queried Barclay mildly, but slightly ironically.

"Not the best," admitted Jennings, "but the only one who knows anything about this apparatus, so you are a liar naturally. But we will want you in about twelve hours to look over the ray apparatus, so you needn't spring that joke about a paralyzer screen." Jennings turned and left. Belasco looked disappointed a moment, then his eyes lighted as he looked at the girl again.

Everyone had been looking at Barclay. Suddenly the elder woman gave a little cry of surprise. "Why! It's Donnie Barclay. I didn't see you at first. How did you get here?" Her voice was deep and powerful.

"Accidentally," replied Barclay with a wry smile, "and unintentionally. They captured my little one-man ship, Mrs. Willman. I'm sorry to meet you again."

Mrs. Willman smiled. "Apparently these pirates want ransom for us. I think Dick will get it. You've recognized Pat haven't you. It's been nearly seven years since you've seen her though." Mrs. Willman didn't seem greatly perturbed.

"I did," nodded Barclay, "as soon as you came in."

"Shut up, you!" growled Belasco, looking at Don. There was a gleam of anticipation in his eyes as he looked at Pat Willman. "And you, little lady, are kinda heavy. But since a damn ---- of a Planetary Patrolman rayed me, I haven't seen a woman. I think it's about time. Take off the things you got on. I wanna see if I like you."

Pat Willman looked at him with unconcealed loathing. "No wonder," she said simply. "And naturally not. And I advise you not to try it."

Belasco grinned down at her from his six-feet-eight, some thirteen inches taller. "I think I will. I'd like it."

Mrs. Willman winked at Barclay, Barclay smiled grimly, and began to weave his fingers into the chains that held him.

Belasco advanced. A grim glint came into the girl's eyes. Belasco's huge paw reached out and grasped her light jacket. With a vicious jerk he pulled it away, tearing the material. Pat Willman stood clad in a light dress. "That," she said with cold anger, "is more than enough."

Belasco grinned, and reached out again. Pat Willman's hand moved like a streak of light, the great wrist was caught in slim white fingers that bit into the flesh, a terrific jerk pulled the giant toward her, and the other hand came up. Belasco was turned around, thrown down, and kicked, thoroughly pounded, then thrown away, gasping and roaring with rage and pain. He was thrown at least fifteen feet across the room, though the ship was moving under an acceleration of an Earth-gravity now. He fell heavily.

He rose with a roar, and rushed back at her. Pat Willman watched him coolly. Pat Willman, twenty-two Earth-years old, had lived practically all her life on Jupiter, a neighbor of Don Barclay. She could move about three times as rapidly as a terrestrial, and had the strength of three terrestrials her own size. She was easily more than a match for any terrestrial man, and she knew it.

CHAPTER IV

Rays and Rays

Barclay had been looking about the room. The guard who had been at the door when the prisoners entered had gone up to his platform, and was interestedly watching Belasco's discomfiture at the hands of the Jovian girl. Belasco had reached her again, and this time she was slapping him. Her slap was comparable to the playful slap of a tiger, and Belasco was getting a more severe beating than many men could have given him. She slapped him with her left hand, which threw him against her right, her right hand slapped his ribs, and hurled him back to the left.

"Keep it up," her mother said softly. "Donnie is doing something."

Pat kept it up, till Belasco's head lolled with each jerk, and blood began to pour from his nose. The guard on the platform didn't interfere, he didn't like Belasco any more than most of the others of the crew. In fact he was laughing. Finally Pat threw the unconscious giant into the middle of the room. The guard looked at him, but made

no move to aid him. Pat Willman also looked at him, her breath coming rapidly, for the air in the ship was thin for such exertion, used as she was to Jupiter's dense atmosphere.

Barclay crouched slowly as she turned to look at him, then, like a flash of light, he bounded straight into the air, and toward the guard platform thirty feet away, and ten up. His first enormous bound carried him directly beneath the shelf-like platform, and as the guard rose with a sudden cry of amazement, to look over the edge, a powerful hand grasped him, and hauled him downward. As he fell, Barclay eased the landing with a catlike softness, his great hand about the throat of the other.

The other hand rose, and with a queer semblance of preciseness, struck the small man on the side of the head. He went suddenly limp and Barclay released him. "Two hours," he said judicially, and leapt over to Belasco. He hit him also on the side of the head with the palm of his hand.

"Three hours, considering what you did to him, Pat," he smiled at the girl. "You took their attention away while I broke those chains. I knew the strength of the chains, and we always have to use double-shackles on Jovians. This is an ex-Patrol ship, and I know a whale of a lot more about it than they do."

Barclay drew Belasco's heat-gun, and his paralyzer gun, gave the heat gun to Pat, and took the paralyzer from the guard. He looked at them hastily. "Hmmm--twenty shots in all, or ten minutes operation. I'll--"

He stopped abruptly as the heavy metal door shot up. Two men stood revealed. They had come for the body of the man Jennings had killed.

With a cry of alarm they sprang back, and brought their paralyzer pistols up. Barclay, moving three times as fast, threw a streaking beam across both of them. Suddenly one of the pistols exploded in a flash of light, then grew gradually red-hot. The guard dropped it as Barclay released him from the beam. In an instant Barclay's swift mind saw what it meant--the pistols were vulnerable to each other, some induction effect released the stored energy of one caught in a ray from another. The dropped pistol was white hot now, smoking and glowing.

From down the corridor came shouts and calls, pounding feet. The reenforcements were coming. "Throw me your pistol!" commanded Barclay, lowering his own ray till it paralyzed only the lower part of the man's body. The second pistol was thrown to him. An instant's glance told him it was full. He slipped it into his belt, and retreated to the platform.

A moment later a crush of pirates tumbled through the still-open door, and came with their paralyzers in their hands. Swiftly Barclay passed his rays over each, and one by one they dropped pistols that were swiftly heating to incandescence.

But from the door, a yet un-rayed pistol reached Barclay, and his weapons were suddenly growing unbearably hot. But they seemed to suck the paralyzing energy into themselves, away from the man, for as he hurled them from him, he drove like a bullet toward the man whose pistol had ruined his weapons. The ray caught him in mid-air, but so swiftly was he coming the man had no time to dodge, and Barclay's

flying body knocked him down, the pistol flying from his hand to crash into wreckage as it struck the metal wall.

Then men began to pile on Barclay. But Barclay was a rubber ball. He bounced about the room with unbelievable speed, his flying hands merely slapping men, but leaving them unconscious for an hour, unless revived by outside aid.

[Illustration: Men began to pile on Barclay. But he was like a rubber ball. He bounced about the room with unbelievable speed, merely slapping men into unconsciousness.]

There were twelve men attacking him, it took him less than forty-five seconds to send them all to sleep.

* * * * *

Now he lost no time. Others were coming from other parts of the ship no doubt, and he had no more pistols.

"Pat--that pistol," he demanded. Without a word the girl tossed it to him. He whirled and shone a beam on a rivet-weld to one side of the heavy metal door. There was a crash, and the great metal slab fell into place as the rivet-weld glowed red-hot under the lashing heat-beam.

"Here's the gun--keep it, and cut your chains. Take care of them, collect their guns and distribute them among the other prisoners here. The door can't be opened from outside nor inside now, till the generators are stopped. This is a last resort for the Patrol men on the ship if mutiny occurs, or prisoners gain control," Barclay smiled broadly. "I'll be back."

He leaped toward the wall where star-studded space looked in through a narrow porthole, set deep in the double walls. He did something to the wall of the room, and a section fell back into the space between the walls. There was air here, air intended for those who might have to make repairs between the walls. But this carefully concealed door was a secret means of escape--and re-capture of the ship. For now, between the walls of the ship, Barclay closed and re-locked the metal door-plate, and made his way along the ribs and braces between the two skins of the ship, climbing against the ship's acceleration toward the bow.

In a minute or so he rested on two cross-beams outside of what he knew was the pilot-room. A mass of cables led from this room, back to the engine-room below. He could have entered the engine-room had he wished, or the ray-room. He preferred the control-room. Every outside room on the ship had these secret panels for emergencies. It was his own idea, and one Commander Morrison hadn't too heartily approved of. But it was proving itself now!

From an almost hidden clamp he extracted a screwdriver, rapidly manipulated it on what appeared to be a signal-junction box, twisted the leads to one side, and then pushed on a bolt-head. There was a sudden, subdued hiss, and a panel sprang backwards as the pneumatic cylinders drove it open. With a single bound, Barclay was in the pilot-house.

Jennings, cold-eyed, bent over the chart table, while Thad stared out of the pilot window, glancing occasionally at the meters and instruments on the board before him. A squat, powerfully built man was working over near the door.

The three turned as one when Barclay entered. Thad, with a trained, dexterous flip of his fingers threw the ship on to automatic control, leaped from his seat, and dove for Barclay. Jennings was whipping a paralyzer pistol from his holster, and the squat man was turning to do battle, his mouth opening for a bull roar.

It seemed like slow motion to the Jovian, while his motions seemed streaks of blurred light to the Earth-trained pirates. Barclay still had the long, sharp-bladed screwdriver, with its heavy insulite handle in his hand. The hand flashed up, and the screwdriver flashed across, to bury itself to the handle in Jennings' wrist. With a scream of pain the pirate-scientist dropped the weapon as the metal blade severed tendons, blood-vessels, and smashed through bone. It had been going with nearly the speed of a pistol bullet, and weighed nearly 12 ounces.

Long before Jennings' pistol could touch the floor, and before Thad had reached him, Barclay grasped the weapon in his hand, whirling in the air as he dove across the small pilot-house. The opened mouth of the squat pirate was emitting sound now, and a pistol was in his hand. Barclay trained his ray-gun on the other in a fraction of a second, and flicked a full-powered beam at it. In an instant a puff of smoke shot out, and the pirate dropped his weapon to see it burst into incandescence as it hit the floor.

From below-decks an answering roar was coming, and the pound of heavy feet. The squat man and Thad reached Barclay almost simultaneously. For some reason Thad had not been wearing any weapons, but now he clung to Don's paralyzer like grim death. The squat man attempted to uppercut to the Patrol Captain's jaw, but Don's head moved a fraction of an inch, and the blow whistled by harmlessly. An open palm struck the side of Thad's head, and the pilot dropped senseless. An instant later the remaining pirate bolted through the door--or attempted to. Barclay caught him, and a moment later he joined Thad on the floor of the pilot-room.

A glance assured Barclay that Jennings had fainted from loss of blood and pain. His hand reached up to a small knob to one side of the entrance-way and a heavy steel panel fell across the one door to the pilot-house, locking in air-tight gaskets. A moment later, and huge magnets had fixed it immovably.

Barclay leaped back to the pilot-board, seated himself swiftly, and threw on the powerful radio set on range 245-671. Swiftly, with nearly 1,000,000 horsepower backing his signals, he sent out a call to Wier, somewhere in near-by space, supporting him in this mad attempt.

"Wier--Wier--P.C. 2500. Wier--Wier--P.C. 2500," he repeated monotonously, listening in his set.

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It became difficult presently as men began to work on the door. Meanwhile his other hand was swiftly altering the course of the ship, bearing back toward the

Jupiter-Mars lane.

"Don--Don--P.C. 2500 answering."

"Temporarily in control here. Heading ship back toward Jupiter-Mars lane. Using spotlight beam and floodlight. Will start radio and leave it going as long as possible. Get direction and distance. Step on it. They're working on this door."

"P.C. 2500 coming," replied the radio set.

Don's flying fingers did things to the radio controls, necessitating considerable work to stop it, and much more work to repair it. But it sent out a continuous high-pitched note. Now he short-circuited certain resistances, and relays, and the ship staggered forward under a full Jovian gravity. There was an abrupt stoppage of work on the door outside, as the terrestrials fell down. Barclay grinned, and locked the controls as they were, adding some bits of wire to hold his shorts in place. Then he got up heavily, and with a tremendous effort walked to the door through which he had entered. From the other side he arranged it to close and lock in place when he released it. Returning, he released it, and watched it merge into the wall.

He returned to the pilot-table once more, seated himself heavily, and listened to Dave Wier's calls.

"Distance 500,000 miles. Approaching rapidly." Carefully Barclay altered the course of the ship's flight till it was pointed toward Wier's machine, and racing toward it even faster. The tremendous acceleration was beginning to tell on the speed indicator.

Barclay glanced back at the door. A dull blue haze was forming about it. "Hmmm--distintegrator didn't stop working. Probably has a range of ten feet, and will hit that wall over there. I wonder--"

Barclay took two heavy bus-bars from the stock room attached to the pilot room, connected them swiftly, bending them over his knee. These he led over, and set up like two poles close together in the necessary path of the disintegration machine's ray. The haze of ionization was growing stronger swiftly. By the time the two bus-bars were in position, there was a tiny pin-prick of blue ray flaming across the room, and a moment later scores of holes appeared.

Then two of them hit the bare copper bus-bars, charged with 150,000 volts. There were two wires. First the door, or what was left of it, exploded into flaming vaporized steel. Next the disintegrator exploded with a terrific concussion, raining white-hot metal parts all about. The machine had short-circuited the twenty-million horsepower generators of the ship, and the result was inevitable.

Barclay looked out through the hole in the door. One man had been badly burned, but the others had crawled laboriously down the hall and away from danger, when they realized that none of them could stand up to turn off the disintegrator and guessed what Barclay, who could stand the weight, would do.

Barclay laughed, and taking his paralyzing pistol, rayed first one then another of the men. Practically the entire crew had come to the defense of the pilot-house. The

others had been working on the brig door. As one after another felt the ray, he slumped, and threw his burning pistol to one side the instant he was released.

Barclay's weapon had two shots left when he finished and returned to the cabin. No sooner had he crossed the threshold than from the ceiling above there poured a rain of the paralyzing rays. His fingers relaxed, and dropped the ray-pistol, which smoked slightly, but didn't explode, as all its charges were gone, his knees held for a moment before he toppled over and fell with a crash.

Joyfully the pirates in the hall hastened to crawl back toward the pilot-house, helping each other. Three made it, and promptly keeled over as they entered the paralyzing rays that filled the room.

Finally one of them crawled back and got a metal bar about five feet long. It took four of them to raise it into position and bring it down on the master-switch, opening it, and instantly shutting off all acceleration.

Then one of the men hauled himself back along the corridor to the engine room, and from the emergency engine-room controls, which were powerless as long as the master-switch was closed, but which now sent the ship ahead under one Earth-gravity.

Gathering a brace of ray pistols from the bench as he ran up to join his friends, he watched them manipulating the metal bar again. It was easier now, and they quickly worked the half-hidden button that had flooded the room with rays. It was just over Jennings' face as he had lain there.

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His eyes were bright and hard as he watched. He was weak, and knew it, holding his right wrist with his left hand. But Barclay wasn't. The shutting off of the paralyzing rays had started him even more quickly than it had started his enemies. With a bound he leaped for them, burying himself in their midst. The metal bar was wrenched from their hands, and thrown across the room. Then the sixteen men who had congregated there got a taste of a Jovian in action. His terrible, flying hands meant certain unconsciousness, or broken bones, while he himself was an invulnerable mass of iron-hard muscle to their blows. The one on the outskirts of the battle who held the two pistols, hesitated to use them, they would paralyze friend and foe alike.

Presently Jennings saw them. "FOOL!" he shrieked over the noise of the fight, "RAY THEM ALL--FOOL!"

The two pistols shot out their beams, and suddenly the fight stopped, Barclay, and those near him fell helpless, those on the outskirts had not been rayed, and with a cheer, fell to pulling their comrades out of the melée.

"He sent a call for help. I couldn't read the radio code, confound him, but he called someone, and got an answer. He shifted our course till we were headed right for that other ship. We were traveling under that acceleration of 4.5 normal for I don't know how long. I'm hurt, the ---- ---- ---- used a screwdriver on me. Threw it like a bullet

from a gun.

"Now: battle stations. Tie that thing up till I'm ready for it. It escaped? Broke its bonds? Then tie it! Put one and a half reverse accelerations, but turn the ship around. First, put that damned radio on the blink, he's got it going, sending a direction-howl. Then make sure the lights are off, and use only the colored lights inside. Check the electrometer."

There was a shrill clangor as the electro-detector caught the tell-tale emanations of powerful atomic generators driven hard. "Ship over-taking us, Captain! Comin' like a streak of light. Funny looking patchwork thing. Must be a thousand years old. I'll bet they plan to ram us, they're comin' too fast to stop," reported the pilot.

"Ray--paralyzer," snapped Jennings.

The generators in the bowels of the ship hummed a deeper note.

"Oh, GOD! Cap--they've _got a screen! The rays don't touch them!_" called the pilot.

Barclay's face lit with a smile, Jennings' eyes opened in consternation.

"Put all your power in induction, switch suddenly to paralyzer," ordered Jennings.

The ray operator's fingers flew. A moment later his face fell. "Didn't touch 'em," he reported.

"Why aren't they raying us?" demanded Jennings in a puzzled tone. "Oh--the prisoners." A smile of satisfaction came over his face. "Listen, throw every kilowatt, even the shield-base power, since they won't ray us, and turn it into the Miller Twin rays, our kind friend installed for us."

Jennings grinned at Barclay as he wrapped a bandage around his injured wrist. The ray operator was working again, and the shrill whine of the generators fell to a rumbling hum. The Twenty-five-Hundred hung motionless beside them now, manned by Jovians it had been able to stop.

A green bulb flashed suddenly on the pilot-board. The Twins were ready.

* * * * *

Two milky tubes of radiance leaped out into space, curved sharply, and cut suddenly across the Twenty-five-Hundred. Something was wrong, the Twin Beams, touching adjacent patches on the ship's side, did not flash into arcing, burning light.

"The paralyzer screen," said Barclay's calm voice in the tense silence, "I helped develop it. You left records when your fiasco demonstration was made."

"Guns," he went on, but his voice was suddenly drowned out. The two patches which the beams touched suddenly flashed with terrific fire. Somewhere below, in the depths of the ship, the great generators rose to a singing whine as the milky beams glowed with blue fires, terrific shattering lightnings at 100,000,000 volts pressure and 15,000,000 horsepower.

They whined like titanic disc saws, then, incredibly they struck some cosmic snag, the shrill whine became an awful, soul-wrenching, nerve-shattering grind, the ship trembled throughout her length, and with an awful abruptness but there was no sound. A man speaking in the engine room suddenly was shouting in silence. The awful calm lasted a fraction of a second. Then it exploded into eye-searing light and terrific concussion. The generators had been re-polarized and exploded, taking a wall with them.

It was quiet again, the faint chug of air-pumps gone.

"Guns," resumed Barclay calmly, "kick--and Miller beams work both ways. I hoped you'd install those beams for me. It was the only way we could destroy your defensive generators without destroying you--to throw them against the hundred million horsepower Ridgely Atomic on that ship."

Jennings stared at him in stupid amazement.

Somewhere behind came the bump and clang as the Twenty-five-Hundred came alongside the helpless ship.

THE END